

Inventory Management System

Overview

This is a comprehensive **Laravel-based Inventory Management System** designed for e-commerce businesses. It provides end-to-end management of products, orders, purchases, warehousing, and financial accounting with deep **eBay marketplace integration**.

Key Features & Modules

1. Inventory Management

- **Product catalog** with SKU, barcode, pricing, and descriptions
- **Product bundles** - composite products with automatic stock calculations
- **Multi-warehouse/rack storage** tracking
- **Barcode generation** and bulk label printing
- **Bulk import/export** via Excel

2. Warehouse & Storage

- Multiple warehouse locations support
- Granular rack/bin level tracking
- Weighted average cost inventory valuation

3. Purchase & Supplier Management

- Purchase order creation and tracking
- Supplier profiles with payables integration
- Partial/full receiving workflows
- Freight and customs duty tracking for imports

4. Sales Channel Integration

- **Full eBay integration** with OAuth2 authentication
- Product listing management across channels
- Real-time inventory sync to marketplaces
- eBay Platform Notifications for webhooks

5. Order Management

- Complete order lifecycle (pending → paid → shipped → delivered)
- Return and refund handling

- **FedEx shipping integration** for labels and tracking
- Shipment deadline alerts

6. Financial Accounting Module

- **Double-entry bookkeeping** system
- Chart of Accounts with hierarchical structure
- Vendor bills and payment tracking
- Automatic journal entries for:
 - Cost of Goods Sold (COGS) on shipment
 - Sales revenue recording
 - Purchase duties and freight
- Reports: Trial balance, ledger, inventory valuation, sales, purchases

7. Dashboard & Reporting

- Customizable widget-based dashboard
- PDF export capabilities
- Multiple report types (sales, purchases, expenses, inventory)

Technology Stack

Layer	Technology
Backend	Laravel (PHP 8.2+)
Database	MySQL/MariaDB
Authentication	Laravel Sanctum
Authorization	Spatie Laravel Permission
Frontend	Blade, Bootstrap 5, Tailwind CSS 4
Build Tool	Vite
PDF Generation	DomPDF
Excel/CSV	Maatwebsite Excel
Barcode	picqer/php-barcode-generator

Database Entities

Core Inventory

- **Products, ProductStock, ProductBundleComponents**
- **Categories, Brands, Warehouses, Racks**

Sales & Orders

- **Orders, OrderItems, OrderMeta**
- **SalesChannels, SalesChannelProduct** (pivot)
- **Shipping**

Purchases

- **Purchases, PurchaseItems, Suppliers**

Accounting

- **ChartOfAccounts, Bills, BillItems**
 - **Payments, JournalEntries, JournalEntryLines**
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Architecture

The application follows a **Service Layer Architecture**:

Controllers → Services → Models → Database

Key services include:

- **BillService, PaymentService, JournalService** (accounting)
 - **InventoryAccountingService** (COGS/revenue automation)
 - **EbayService, EbayOrderService, EbayApiClient** (eBay integration)
 - **FedexService** (shipping)
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Integration Workflows

eBay Flow

OAuth2 Auth → Subscribe Notifications → Import Listings → Sync Inventory

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Receive Order Webhooks → Create Local Orders → Ship → Handle Returns

Accounting Flow

Purchase (with duties/freight) → Receive Stock → Update Valuation

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Customer Order → Ship (COGS entry) → Record Revenue → Reconcile

Console Commands

- `ebay:sync-orders` - Poll eBay for orders
 - `ebay:sync-listings` - Import listings as products
 - `ebay:refresh-tokens` - Refresh OAuth tokens
 - `check:delivery-status` - Check shipment status
 - `shipping:refresh-tokens` - Refresh FedEx tokens
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Summary

This is a **production-ready, enterprise-grade inventory system** ideal for e-commerce sellers, particularly those on eBay, who need integrated inventory management, order fulfillment, and financial accounting in a single platform.

LinkedIn Post Content:

Excited to share a project I've been working on! 🚀

I recently built a complete Inventory Management System from the ground up using Laravel, and I wanted to share what went into it.

What does it do?

It's an all-in-one platform for e-commerce businesses that handles:

→ Product & inventory tracking across multiple warehouses → Purchase orders and supplier management
→ Order processing with full lifecycle management → Complete double-entry accounting system → eBay marketplace integration with real-time sync → FedEx shipping integration for labels & tracking

The accounting module was probably the most challenging part - building a proper chart of accounts, automatic journal entries for COGS, and generating financial reports like trial balance and ledger statements.

The eBay integration was another interesting challenge. It handles OAuth2 authentication, inventory sync, order webhooks, and even return/refund processing automatically.

Tech stack:

- Laravel (PHP 8.2+)
- MySQL
- Bootstrap 5 & Tailwind CSS
- Vite for frontend builds
- Queue-based background processing

Some features I'm particularly proud of:

- Bundle products that automatically calculate available stock based on components
- Weighted average cost inventory valuation
- Real-time inventory sync across sales channels
- Automated accounting entries when orders ship

Building this taught me a lot about handling complex business logic, integrating with third-party APIs, and designing systems that actually solve real problems for businesses.

Would love to connect with others working on similar projects or anyone in the e-commerce/inventory space!

#Laravel #PHP #WebDevelopment #Ecommerce #InventoryManagement #SoftwareEngineering
#Programming #FullStackDeveloper